Samrath Bawa

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Education

California State Polytechnic University, Pomona — Pomona, CA
Bachelor of Science in Computer Science, GPA 3.61, Dean's List — *2025 – 2029*

- Coursework: Data Structures and Algorithms, Computer Organization and Assembly, Systems Programming, C++, Discrete Structures, UI Design, Probability and Statistics

Technical Skills

Languages: C++, Java, Python, R, JavaScript, HTML, CSS, x86 Assembly
Frameworks and Libraries: pandas, JUnit, REST APIs
Developer Tools: Git, PlatformIO
Embedded and Electrical: Teensy 4.1, ESP32, CAN bus, Altium Designer, KiCad
CAD and Analysis: SolidWorks, Fusion 360, Ansys Mechanical (FEA)
Fabrication: 3D printing, CO2 laser cutting and engraving, woodworking

Experience

Software Captain — Oct 2026 – Present
Bronco Electric Motorsports (Formula SAE Electric) — *Pomona, CA*

- Lead a software team of 7 on the electric race car and set its code, **Git**, and documentation standards
- Develop the car's firmware in **C++** for **Teensy 4.1** and **ESP32** microcontrollers using **PlatformIO**, and own the **CAN bus** network linking the dashboard, wheel display, and sensor board
- Review team members' pull requests to enforce the team's code standards
- Refactored last year's full firmware codebase into modules that the new car is built on

AI Training Fellow — Jul 2026 – Present
Handshake AI — *Remote*

- Design hard questions about charts, tables, and images to find where AI models break, tagging each by the kind of reasoning it tests
- Author money and tax questions that can only be answered by reading through five or more documents
- Build grading rubrics that score AI answers point by point, and document what each answer got wrong as training data
- Rate and caption AI generated video, images, and audio against a detailed style guide

Projects

Race Car Dashboard and Steering Wheel Display | *Teensy, CAN bus, Altium*

- Designed a sunlight readable **CAN** wheel display and a **Teensy 4.1** dashboard schematic for the car
- Migrated the team's **ESP32** sensor board from **KiCad** to **Altium**, verifying all 110 parts and 75 nets

Accelerator Pedal Stress Analysis | *SolidWorks, Ansys*

- Ran **FEA** on the pedal shock mounts, found the stock part too close to failing, and designed an aluminum pedal stop (safety factor 2.4)
- Built a posable 95th percentile driver model in **SolidWorks** to check the car against the FSAE driver fit rules

Racing Game Force Feedback Bridge | *Windows, Device Drivers*

- Converted live racing game data into steering forces on a real wheel using **virtual device drivers**

Leadership and Volunteering

Vice President — Aug 2023 – Jun 2025
Woodbridge High School Web Development Club

- Taught members HTML, CSS, and JavaScript and built websites for clients such as other clubs and teachers

Jr. Auxiliary Volunteer — May 2024 – May 2025
Hoag Hospital

- Supported patients, visitors, and staff with wayfinding and wheelchair services